Thermodynamic description of the Ag–Cu system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ag–Cu binary system as calculated with pycalphad from the SGTE dataset AgCu-06He.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Ag) + (Cu), lies at 1055.8 K.

1. Introduction

The Ag–Cu system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of X.C. He, H. Wang, H.S. Liu, Z.P. Jin, Calphad, 30, 367-74(2006). The calculations use pycalphad 0.11.2 at 1 atm.

Pure Ag melts at 1234.9 K. For Cu, the liquidus meets the axis at 1357.8 K.

2. Phase diagram

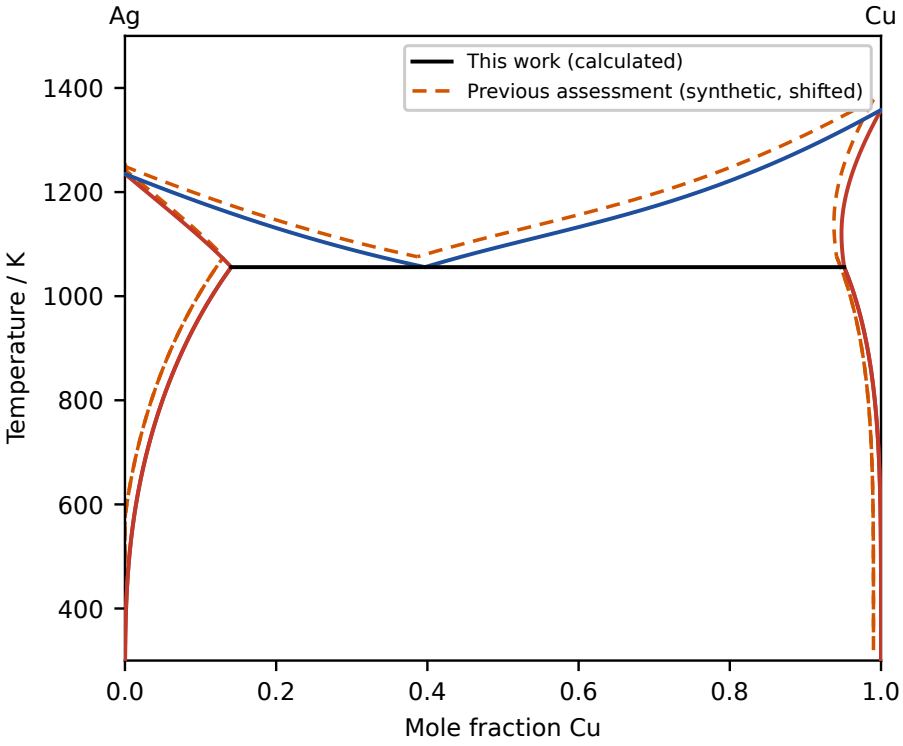


Fig. 1. Calculated Ag–Cu phase diagram and a dashed, deliberately shifted second assessment.

At 1055.8 K the eutectic L → (Ag) + (Cu) takes place; the L phase then contains 39.7 at.%.

Table 1. Invariant reactions in the Ag–Cu system (calculated).

Reaction	Type	T (K)	x(Cu), 1	x(Cu), 2	x(Cu), 3
L → (Ag) + (Cu)	eutectic	1055.8	0.397	0.141	0.951

3. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its maximum, 3683 J/mol, at 54.0 at.% Cu.

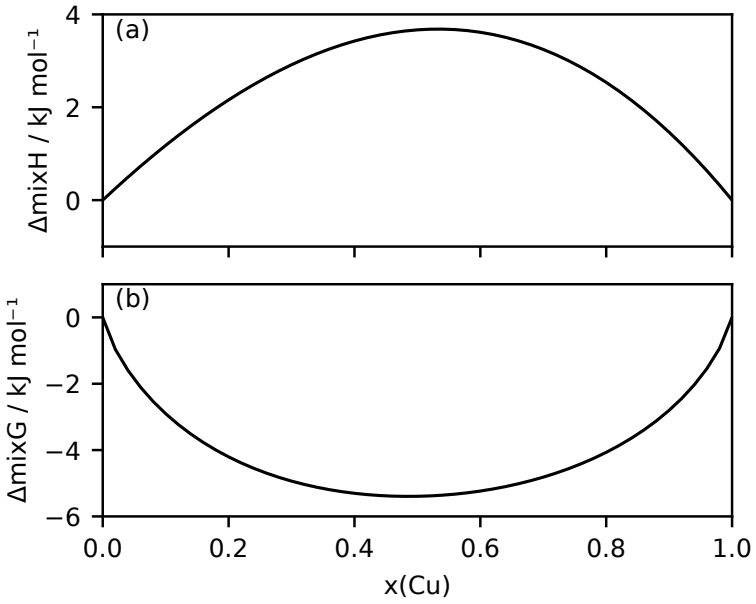


Fig. 2. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1450 K.

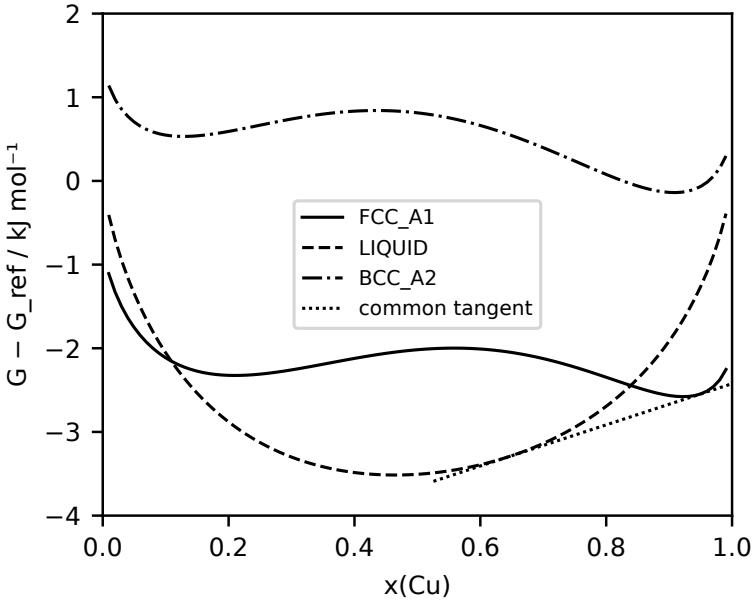


Fig. 3. Calculated Gibbs energy curves at 1150 K.

Table 2. Redlich–Kister parameters of the liquid (Ag–Cu).

ν	a / J mol⁻¹	b / J mol⁻¹ K⁻¹	c / J mol⁻¹ K⁻¹
L0	16914.949	-14.7721	1.54955
L1	-1963.3	0.8623	0.0

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.